
When the price tells the truth

One battery, two hours, and the value of flexibility

A minimal example, with two generators and one ramp constraint, showing where a battery's revenue comes from and why it is exactly equal to what the battery saves the system. Then the same reading on a real day of the Brazilian system.

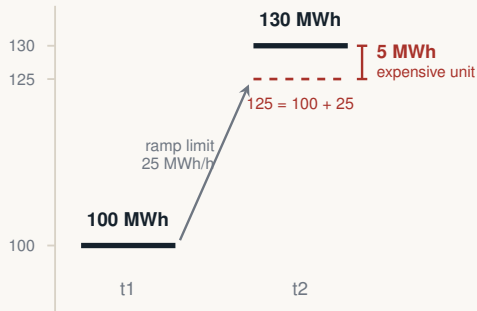
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THE PROBLEM

Demand rises 30 MWh in one hour. The cheap unit rises 25.

- Two thermal units, unlimited capacity.
- Variable cost of **100** and **300** R\$/MWh.
- Ramp-up limit of **25 MWh/h**, the same for both.
- Demand of **100 MWh** at t1 and **130 MWh** at t2.

t1 is the first hour of the horizon, so the ramp constrains only the step from t1 to t2.



Vertical axis truncated at 90 MWh.

CASE A • NO BATTERY

The optimal dispatch costs R\$ 24,000, and the price at t1 turns negative.

	t1	t2	Revenue	Cost	Profit
Demand	100	130	−29,000		
Thermal 1	100	125	27,500	22,500	5,000
Thermal 2	0	5	1,500	1,500	0
Hourly cost	10,000	14,000			
Spot price	−100	300			
Total			0	24,000	5,000

Quantities in MWh; revenue, cost, profit and hourly cost in R\$; spot price in R\$/MWh. Demand's revenue enters negative, because it is what consumers pay. Each hour's spot price is the marginal operating cost: the change in total cost over the two periods when 1 MWh of demand is added in that hour and the other one is held fixed.

The dispatch model

$$\min_{g \geq 0} \sum_t \sum_i c_i g_{i,t}$$

subject to

$$\sum_i g_{i,t} = d_t \quad (\lambda_t)$$

$$g_{i,2} - g_{i,1} \leq R \quad (\mu_i)$$

λ_t is the price of energy in hour t . μ_i is what one more MWh of ramp is worth to the system.

The negative price is not a flaw in the model. It is the right arithmetic.

$$\mu_1 = c_2 - c_1 = 200$$

$$\lambda_1 = c_1 - \mu_1 = -100$$

$$\lambda_2 = c_1 + \mu_1 = +300$$

$$\lambda_2 - \lambda_1 = 2(c_2 - c_1) = 400$$

- Ramp headroom is worth the difference between the two units.
- Generating at t1 costs 100 and *creates* headroom worth 200. Net, -100.
- Generating at t2 costs 100 and *consumes* headroom worth 200. Total, 300.
- The spread between the hours is twice the cost difference.

Revenue adequacy

Consumers pay 29,000 and generators receive 29,000. The revenue column sums to zero.

Cost recovery

Thermal 1 sells at -100 at t1 and still closes with a profit of 5,000, which is exactly $\mu_1 \times R = 200 \times 25$, the rent on its own ramp. Thermal 2 closes at zero. No generator loses money.

Electricity has no waste bin. Since it is impossible to generate more than is consumed, the negative price at t1 is the system saying that consuming in that hour is worth 100 R\$/MWh.

CASE B • WITH A 1 MWh BATTERY

The battery buys at -100 , sells at $+300$, and earns 400.

	t1	t2	Revenue	Cost	Profit
Demand	100	130	-29,000		
Thermal 1	101	126	27,700	22,700	5,000
Thermal 2	0	3	900	900	0
Battery	-1	+1	400	0	400
Hourly cost	10,100	13,500			
Spot price	-100	300			
Total			0	23,600	5,400

The battery charges 1 MWh at t1, when the price is -100 , and is therefore paid 100 to consume. It discharges that same 1 MWh at t2 and sells at 300. Revenue of 400 and zero cost. Prices do not change, because the units at the margin are still the same.

The model with the battery

$$\min_{g, e \geq 0} \sum_t \sum_i c_i g_{i,t}$$

subject to

$$\sum_i g_{i,1} = d_1 + e \quad (\lambda_1)$$

$$\sum_i g_{i,2} + e = d_2 \quad (\lambda_2)$$

$$g_{i,2} - g_{i,1} \leq R \quad (\mu_i)$$

$$e \leq E \quad (\beta)$$

e is the energy shifted from t1 to t2, with $E = 1$ MWh and 100% efficiency. The dual of the capacity limit is the price difference, $\beta = 400$.

THE TWO CASES SIDE BY SIDE

The battery's profit is the saving it creates for the system.

	Without battery			With battery		
	Revenue	Cost	Profit	Revenue	Cost	Profit
Thermal 1	27,500	22,500	5,000	27,700	22,700	5,000
Thermal 2	1,500	1,500	0	900	900	0
Battery				400	0	400
Consumers	−29,000			−29,000		
Total	0	24,000	5,000	0	23,600	5,400

Operating saving
24,000 → 23,600
−400 R\$

Battery profit
400 − 0
+400 R\$

Paid by consumers
29,000 = 29,000
unchanged

The battery's income is a scarcity rent, and it vanishes at 2.5 MWh.

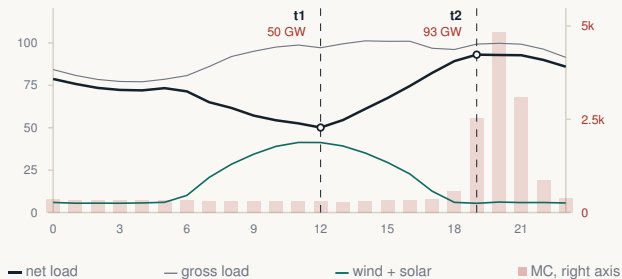
Installed battery	Total cost	Price t1	Price t2	Marginal value
0 MWh	24,000	-100	300	400
1 MWh	23,600	-100	300	400
2 MWh	23,200	-100	300	400
2.5 MWh or more	23,000	100	100	0

Total cost in R\$; prices and marginal value in R\$/MWh. The marginal value is what the next MWh of battery is worth to the system.

With 2.5 MWh of storage the ramp stops binding, the expensive unit leaves the dispatch, and both hours settle at 100 R\$/MWh. From there on, the next MWh of battery is worth zero.

The same price that pays the first battery tells us when the next one stops being needed. That is how price connects the market, and the incentives it creates, to the needs of those who operate the system.

The fleet climbs 43 GW while MC multiplies by 8.

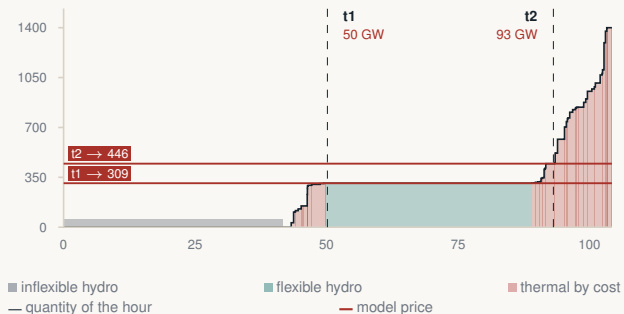


t1 is the trough (12:00) and t2 the peak (19:00) of the controllable fleet curve. Gross load barely moves: what climbs 43 GW is the fleet.

	t1 · 12:00	t2 · 19:00	Δ
Gross load	97,168	99,352	+2,185
Wind + solar	41,281	5,464	−35,816
Distributed	28,428	278	−28,150
Hydro	41,696	81,048	+39,352
Thermal	8,453	12,064	+3,611
Controllable	50,148	93,111	+42,963
MC (SE)	309	2,521	×8.2
Spot (SE)	350	1,200	×3.4

Two days later the same system moved 27 GW and MC moved 47 R\$/MWh. On the 4th it moved 43 GW and MC moved 2,212. The value of flexibility is concentrated in a handful of hours of the year.

The hydro ramp is what separates the two prices.



The dashed verticals are quantities, what the fleet had to deliver in each hour; the horizontals are prices. They pull apart when flexible hydro can no longer reach the peak. Ramp of 14.3%/h of flexible hydro, the day's own.

R\$ m	Without bat. 309 → 446			With bat. 309 → 347		
	Rev.	Cost	Profit	Rev.	Cost	Profit
Inflex. hydro	31.5	0.0	31.5	27.3	0.0	27.3
Flexible hydro	17.7	12.3	5.4	14.4	12.9	1.5
Thermal	7.9	4.3	3.6	6.0	3.4	2.6
Battery				0.1	0.0	0.1
Net load	-57.0			-47.8		
Total	0.0	16.5	40.5	0.0	16.3	31.5

With 2,000 MWh of battery. Revenue sums to zero in both cases.

Demand pays R\$ 9.2 million less, and operating cost falls only R\$ 0.2 million: most of it is a transfer from inframarginal generators to consumers, and the battery keeps R\$ 0.1 million.